PROBLEM PROJECT SUBMISSION

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Group 3 Project

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Description of the Real-World Problem

Municipalities are responsible for maintaining important public services and infrastructure, such as roads, streetlights, water systems, drainage, and waste management. Residents often need to report problems such as potholes, broken streetlights, water leaks, burst pipes, illegal dumping, blocked drains, sewage problems, damaged roads, and electricity infrastructure issues.

A problem is that residents may not always have an easy way to report these issues and track what happens after a complaint has been submitted. Complaints may also be difficult for municipal staff to organise, prioritise, assign to technicians, and monitor until completion.

The people affected include residents, municipal employees, maintenance teams, and the wider community. The problem is important because delays in identifying and resolving infrastructure issues can negatively affect public safety, service delivery, and the quality of the local environment.

Proposed Software Solution

We propose developing a Municipal Problem Reporting & Tracking System, which would allow residents to submit municipal service complaints through a web or mobile-based system.

When submitting a complaint, a resident could provide information such as:

- Type of problem
- Description of the problem
- Location
- Photo or other supporting evidence
- Date reported

The system would generate a unique reference number, for example REQ-10482, which the resident could use to track the complaint.

The complaint could move through statuses such as:

Submitted → Assigned → In Progress → Resolved → Closed

Residents would be able to view the status of their complaint, the assigned technician where appropriate, and an estimated completion date.

Municipal staff would have access to a dashboard showing information such as:

- Open complaints
- High-priority complaints
- Complaints by area

- Average resolution time
- Technician workload
- Frequently reported or repeat problems

The system could also support notifications to residents when their complaint status changes.

Why Our Group Chose This Problem

Our group chose this problem because municipal service delivery is an important issue within South African communities, and we believe that software can be used to improve the way residents communicate problems to their municipality.

We are interested in investigating this problem because it provides an opportunity to develop a practical system that addresses a real-world need rather than creating a system for a purely hypothetical situation.

The project will also allow our group to apply the programming, database, web development, and software engineering concepts that we have learned. Since the system involves different types of users, permissions, workflows, databases, reporting, and business logic, it provides a suitable challenge for our group while remaining achievable within the scope of the project.